

$$\begin{array}{l}
 \text{Row 3} = \text{Row 3} \\
 \times 1/3 \rightarrow \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} + 2 \times \text{Row 3}} \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} + \text{Row 2}} \begin{bmatrix} 1 & 4 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{array}$$

By the Invertible Matrix Theorem and $\Rightarrow A$ is an invertible matrix. $\square$

8. $\begin{bmatrix} 1 & 3 & 7 & 4 \\ 0 & 5 & 9 & 6 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 10 \end{bmatrix}$

$$\begin{array}{l}
 \text{Row 4} = \text{Row 4} \\
 \times 1/10 \rightarrow \begin{bmatrix} 1 & 3 & 7 & 4 \\ 0 & 5 & 9 & 6 \\ 0 & 0 & 2 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row 3} = \text{Row 3} \times 1/2} \begin{bmatrix} 1 & 3 & 7 & 4 \\ 0 & 5 & 9 & 6 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} - 9 \times \text{Row 3}} \begin{bmatrix} 1 & 3 & 7 & 4 \\ 0 & 5 & 0 & -30 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} - 7 \times \text{Row 3}} \begin{bmatrix} 1 & 3 & 0 & -24 \\ 0 & 5 & 0 & -30 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{array}$$

$$\begin{array}{l}
 \text{Row 2} = \text{Row 2} \\
 \times 1/5 \rightarrow \begin{bmatrix} 1 & 3 & 0 & -24 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} - 3 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{array}$$

By the Invertible Matrix Theorem and $b \neq 0$, A is an invertible matrix. $\square$

In Exercises 11 and 12, the matrices are all $n \times n$. Each part of the exercises is an implication of the form "If 'statement 1', then 'statement 2'".

Mark an implication as True if the truth of "statement 2" always follows whenever "statement 1" happens to be true. An implication is False if there is an instance in which "statement 2" is false but "statement 1" is true.

Justify each answer.

11) a) If the equation $Ax=0$ has only the trivial solution, then A is row equivalent to the $n \times n$ identity matrix.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (a) and (b), we know that the given